PHYSIOLOGY · MCQ PRACTICE

GUE BLOCK

Water & Electrolyte Balance by the Kidneys — MCQ Practice

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HOW TO USE THIS SET

20 questions, 5 options each (A–E). Answer on paper or in your head, then check every answer against the key on the final page.

4 Easy · 10 Medium · 6 Hard · suggested time ≈ 25 minutes.

Covers water balance, fluid compartments, nephron segment handling, ADH, RAAS/aldosterone, and potassium balance.

Q Questions

1. Roughly what percentage of total body water is intracellular fluid (ICF)? EASY

A. ⅓

B. ½

C. ⅔

D. 90%

E. 10%

2. What is the major extracellular cation? EASY

A. K^+

B. Na^+

C. Ca^{2+}

D. Mg^{2+}

E. HCO_3^-

3. Which nephron segment does ADH act on to increase water reabsorption? EASY

A. Proximal convoluted tubule

B. Descending limb of the loop of Henle

C. Thick ascending limb

D. Collecting duct

E. Bowman's capsule

4. Approximately what proportion of daily water output leaves via urine? EASY

A. ≈100 mL/day

B. ≈400 mL/day

C. ≈500 mL/day

D. ≈1,500 mL/day

E. ≈2,500 mL/day

5. Which nephron segment reabsorbs Na^+, K^+, and $2Cl^-$ via the NKCC2 transporter while remaining impermeable to water? MEDIUM

A. Proximal convoluted tubule

B. Descending limb of the loop of Henle

C. Thick ascending limb of the loop of Henle

D. Distal convoluted tubule

E. Collecting duct

6. A person drinks several liters of water. What happens to ADH secretion and urine output? MEDIUM

A. ADH rises; urine becomes concentrated and low-volume
B. ADH falls; urine becomes dilute and high-volume
C. ADH is unaffected by water intake
D. Urine volume falls regardless of ADH
E. Aldosterone alone determines the response, not ADH

7. Which hormone's distal-nephron actions include both increasing Na^+ reabsorption AND increasing K^+ secretion? MEDIUM

A. ADH
B. Aldosterone
C. ANP
D. PTH
E. Renin

8. What is the trigger for renin release from juxtaglomerular cells? MEDIUM

A. High NaCl delivery to the macula densa
B. Low renal perfusion / low NaCl delivery to the macula densa
C. High atrial stretch
D. High plasma potassium only
E. High ADH levels

9. In the medullary countercurrent gradient, which statement is correct? MEDIUM

A. Cortical osmolality is higher than inner medullary osmolality
B. Osmolality rises from ≈300 mOsm/kg in the cortex to as high as ≈1,200 mOsm/kg in the inner medulla
C. The gradient is built entirely by the proximal tubule
D. The vasa recta destroy the gradient
E. Osmolality is uniform throughout the medulla

10. Which statement correctly distinguishes total-body Na^+ content from plasma $[Na^+]$ concentration? MEDIUM

A. They are the same thing and always change together
B. Total-body Na^+ mainly affects ECF volume; plasma $[Na^+]$ mainly reflects water balance relative to solute
C. Total-body Na^+ only affects ICF volume
D. Plasma $[Na^+]$ is set entirely by potassium balance
E. Total-body Na^+ has no relationship to blood pressure

11. Which of the following increases DCT calcium reabsorption? MEDIUM

A. Aldosterone
B. ADH
C. Parathyroid hormone (PTH)
D. Atrial natriuretic peptide
E. Renin

12. What is the expected renal response to atrial stretch from volume overload? MEDIUM

A. Renin release increases
B. ANP is released, promoting natriuresis (Na^+ excretion)
C. ADH release is suppressed to zero permanently
D. Aldosterone release increases to retain more Na^+
E. No hormonal response occurs

13. A patient has a selective lesion destroying collecting-duct principal cells' V_2 receptors, with the rest of the nephron intact. What is the most likely consequence? MEDIUM

A. Impaired water reabsorption despite normal or high ADH levels, since the receptor needed to insert aquaporin-2 is lost

B. Increased aldosterone secretion to compensate directly for the ADH defect

C. No consequence, since V2 receptors are not involved in water reabsorption

D. Loss of Na^+ reabsorption only, water handling unaffected

E. Complete loss of proximal tubule glucose reabsorption

14. Which best explains why potassium balance is regulated primarily by varying secretion rather than varying filtration or early reabsorption? MEDIUM

A. Most filtered K^+ is reabsorbed earlier in the nephron, so the late distal tubule/collecting duct secretion is the key adjustable step

B. K^+ is never filtered at the glomerulus

C. K^+ reabsorption is fixed and cannot be regulated anywhere

D. Aldosterone has no effect on K^+ handling

E. K^+ secretion only happens in the proximal tubule

15. A patient with central diabetes insipidus has no endogenous ADH. A trial of desmopressin (a V_2 agonist) is given. If this patient instead had nephrogenic diabetes insipidus (V_2 receptors non-functional), what would distinguish the two conditions on this treatment trial? HARD

A. Both conditions respond identically to desmopressin, since both lack functioning ADH signaling

B. Central DI responds to desmopressin with concentrated urine (receptor intact, ligand restored); nephrogenic DI does not respond, since the receptor itself is defective

C. Nephrogenic DI responds better because it has higher baseline ADH

D. Desmopressin works only in nephrogenic DI

E. Neither condition involves the V2 receptor pathway at all

16. A patient has a selective defect in NKCC2 function in the thick ascending limb, with the rest of the nephron intact. Predict the effect on the kidney's ability to produce maximally concentrated urine during dehydration. HARD

A. Concentrating ability is unaffected, since NKCC2 only matters for dilute urine

B. Concentrating ability is impaired, because NKCC2-driven NaCl reabsorption in the water-impermeable TAL is what builds the medullary osmotic gradient ADH-driven water reabsorption depends on

C. Concentrating ability improves, since more water stays in the ascending limb

D. Only proximal tubule glucose reabsorption would be affected

E. ADH secretion would compensate completely regardless of the gradient

17. Two patients present with low effective circulating volume. Patient A has an intact renin-angiotensin-aldosterone axis; Patient B has primary adrenal insufficiency (no aldosterone production) but an otherwise intact axis up to angiotensin II. Compared to Patient A, Patient B would be expected to show: HARD

A. Normal Na^+ reabsorption and K^+ secretion, since angiotensin II alone suffices

B. Impaired distal Na^+ reabsorption and impaired K^+ secretion, since aldosterone is the final effector acting on ENaC and Na^+/K^+-ATPase at the distal nephron

C. Excess K^+ secretion due to unopposed angiotensin II

D. No difference, since renin release is independent of aldosterone

E. Complete absence of any RAAS activation

18. A patient retains excess total-body sodium due to high dietary intake and impaired renal excretion, but drinks enough water that plasma [Na^+] concentration stays within the normal range. Which statement best reflects the physiology described in the lecture? HARD

A. This is impossible — total-body Na^+ and plasma [Na^+] must always move together

B. Total-body Na^+ excess is expanding ECF volume (and likely raising blood pressure) even though plasma [Na^+] concentration — which mainly reflects water relative to solute — remains normal

C. Normal plasma [Na^+] rules out any Na^+ balance problem

D. This pattern only occurs with simultaneous K^+ depletion

E. ECF volume is determined by potassium, not sodium, in this scenario

19. A patient has isolated loss of aquaporin-2 trafficking (a signaling defect downstream of the V2 receptor, so the receptor itself binds ADH normally) but RAAS and aldosterone signaling are fully intact. During water deprivation, which of the following is most likely? HARD

A. Normal urine concentration, since aldosterone compensates for any ADH pathway defect

B. Impaired urine concentration despite elevated ADH, because aquaporin-2 insertion — not aldosterone — is what increases collecting-duct water permeability

C. Increased GFR fully corrects the defect

D. The defect would instead cause excess water reabsorption

E. No effect, since AQP2 only matters in the proximal tubule

20. Comparing the proximal tubule and the collecting duct: both reabsorb Na^+ and water, yet only the collecting duct's water reabsorption is described as hormonally variable (via ADH). What best accounts for this difference, based on the lecture's description of each segment? HARD

A. The PCT reabsorbs a fixed ≈65% of Na^+ and water as a bulk, continuous process, while the collecting duct's water reabsorption is explicitly described as variable and ADH-dependent — making the collecting duct the site of fine, physiologically adjustable control

B. The PCT's Na^+ and water reabsorption is itself ADH-dependent

C. The collecting duct does not actually reabsorb water

D. Both segments are equally hormonally regulated

E. The distinction is not meaningful physiologically

✓ Answer key

Q	Answer	Difficulty	Why
1	**C**	EASY	ICF is approximately ⅔ of total body water; ECF is approximately ⅓.
2	**B**	EASY	Na^+ is the major ECF cation; K^+ is the major ICF cation.
3	**D**	EASY	ADH binds V_2 receptors on collecting-duct principal cells, inserting aquaporin-2 channels there.
4	**D**	EASY	Of the ≈2,500 mL/day total output, urine accounts for ≈1,500 mL/day, the largest single route.
5	**C**	MEDIUM	The thick ascending limb reabsorbs $Na^+/K^+/2Cl^-$ via NKCC2 and is impermeable to water — central to building the medullary gradient.
6	**B**	MEDIUM	Water loading lowers plasma osmolality, suppressing ADH; fewer aquaporin-2 channels are inserted, producing a large volume of dilute urine.
7	**B**	MEDIUM	Aldosterone increases ENaC and Na^+/K^+-ATPase activity at the distal nephron/collecting duct, producing both effects together.
8	**B**	MEDIUM	Reduced renal perfusion or reduced NaCl delivery to the macula densa triggers renin release, starting the RAAS cascade.
9	**B**	MEDIUM	The gradient rises from ≈300 mOsm/kg (cortex) to up to ≈1,200 mOsm/kg (inner medulla), preserved by the vasa recta and built by the loop of Henle.
10	**B**	MEDIUM	This is an explicit distinction in the lecture: total-body Na^+ drives ECF volume/pressure, while plasma [Na^+] concentration mainly reflects water balance relative to solute.
11	**C**	MEDIUM	The DCT reabsorbs Ca^{2+}, a process increased by PTH, as distinct from Na^+ handling (NCC) in the same segment.
12	**B**	MEDIUM	

			Atrial stretch triggers ANP release, which generally promotes natriuresis — opposing the Na^+-retaining action of RAAS.
13	**A**	MEDIUM	ADH must bind the V_2 receptor to trigger aquaporin-2 insertion; destroying the receptor blocks the water-reabsorption response even if ADH itself is present.
14	**A**	MEDIUM	Most filtered K^+ is reabsorbed earlier in the nephron; the late distal tubule and collecting duct provide the variable, aldosterone-sensitive secretion that fine-tunes K^+ balance.
15	**B**	HARD	Central DI is a deficiency of ADH ligand with an intact receptor — supplying desmopressin (a V2 agonist) restores water reabsorption. Nephrogenic DI is receptor-level resistance, so supplying more ligand cannot produce a response, distinguishing the two.
16	**B**	HARD	The TAL's NaCl reabsorption via NKCC2, combined with its water impermeability, is what establishes the medullary osmotic gradient; without it, even maximal ADH/aquaporin-2 activity in the collecting duct has a weaker gradient to pull water into, impairing maximal urine concentration.
17	**B**	HARD	Aldosterone is the final effector of the RAAS cascade at the distal nephron, driving ENaC and Na^+/K^+-ATPase activity (↑Na^+ reabsorption, ↑K^+ secretion). Without aldosterone, upstream angiotensin II generation does not translate into this distal tubular response.
18	**B**	HARD	The lecture draws an explicit distinction: total-body Na^+ content drives ECF volume (and blood pressure) independent of plasma [Na^+] concentration, which instead tracks water balance relative to solute — so Na^+ retention with proportional water retention can expand ECF volume while plasma [Na^+] stays normal.
19	**B**	HARD	Aldosterone governs Na^+ reabsorption/K^+ secretion, not water permeability. Water reabsorption in the collecting duct depends specifically on ADH-driven aquaporin-2 insertion; a defect downstream of the receptor in that pathway cannot be rescued by an intact RAAS axis.
20	**A**	HARD	The lecture describes the PCT as reabsorbing a set ≈65% of filtered Na^+ and water (bulk, relatively fixed reabsorption), while collecting-duct water reabsorption is explicitly labeled variable and increased by ADH — the site where final urine concentration is tuned to the body's needs.